CH 5 — Differentiability

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Recall $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x,y)\neq 0 \\ 0 & (x,y)=(0,0) \end{cases}$

We have previously shown that f is not continuous at $(0,0)$

But $f_x(0,0) = f_{y(0,0)} = 0$. Thus partial derivatives exists

Notice that here differentiability does not imply continuity

Recall in Calculus I where cusps, jump discontinuity, vertical asymptotes are not differentiable

When $f(x)$ differentiable, the linear approximation is defined and is a good local approximation on the graph

> 📘 **Definition — Remainder Approximation**
>
> If $g'(a)$ exists, then
>
> $$\lim_{x\to a} \frac{|g(x) - L_a(x)|}{|x-a|} = \lim_{x\to a} \frac{|R_{1,a}(x)|}{|x-a|}$$

Proof

$R_{1,a}(x) = g(x) - L_a(x) = g(x) - [g(a) + g'(a)(x-a)]$

$\frac{|R_{1,a}(x)|}{|x-a|} = \frac{|\ g(x)-[g(a)+g'(a)(x-a)]|}{|x-a|} = |\frac{g(x)-g(a)}{x-a} - g'(a)|, \quad x \neq a$

$\lim_{x\to a} \frac{|R_{1,a}(x)|}{|x-a|} = \lim_{x\to a} |\frac{g(x)-g(a)}{x-a} - g'(a)| = |g'(a) - g'(a)| = 0$

□

> 📘 **Definition — Differentiability**
>
> $f(x)$ is differentiable at $x = a$ if
>
> $$\lim_{x-a} \frac{|R_{1,a}(x)|}{|x-a|} = 0$$
>
> $f(x,y)$ is differentiable at $(x,y) = (a,b)$ if
>
> $$\lim_{(x,y)\to(a,b)} \frac{|R_{1,a}(x,y)|}{\|(x,y)-(a,b)\|} = 0$$

Note: for the linearization to exist at (a,b), both partial derivatives must exists at (a,b) but this is not enough to guarantee differentiability. The existence of the partial derivatives is a necessary condition, but not sufficient condition for differentiability.

Example:

Show that $g(x) = x^2$ satisfy the above definition on differentiability at $x = 1$.

$R_{1,a} = g(x) - [g(a) + g'(a)(x - a)] = x^2 - 1 - 2(x - 1) = x^2 - 2x + 1$

$\lim_{x\to 1} \frac{|x^2-2x+1|}{|x-1|} = \lim_{x\to a} |\frac{x^2-2x+1}{x-1}| = \lim_{x\to a} |x - 1| = 0$

Definition — Derivatives at a Point

If $f(x, y)$ satisfy

$$\lim_{(x,y)\to(a,b)} \frac{|f(x, y) - f(a, b) - c(x - a) - d(y - b)|}{|(x, y) - (a, b)|} = 0$$

then $c = f_{x(a,b)}, d = f_{y(a,b)}$